Secondary School Exam Pass Rates and Performance Analysis by Subject and State

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Problem Overview

Secondary school examination outcomes reflect structural educational disparities across geographical jurisdictions, curricula frameworks, and disciplinary verticals. This project implements an analytical pipeline to examine student pass rates across secondary boards and states in India, evaluating STEM versus Humanities performance gaps, isolating gender-stratified achievement differentials, handling missing examination audit records, and visualizing cross-state trajectories.

Dataset Reference

- **Source:** Open Government Data (OGD) Platform India / Ministry of Education
- **Direct Resource Link:** Secondary School Exam Performance Dataset (data.gov.in)

Technical Environment

- **Runtime Environment:** Google Colab (Python 3.10+)
- **Core Processing Libraries:** `numpy` (array arithmetic & numerical functions), `pandas` (tabular transformation & aggregations)
- **Visualization Engines:** `matplotlib.pyplot` (canvas layout & rendering), `seaborn` (statistical chart distributions)

```
# ============================================================================
# SECTION 1: ENVIRONMENT SETUP & LIBRARY IMPORT
# ============================================================================
import os
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Visual configuration
sns.set_theme(style="whitegrid", palette="muted")
plt.rcParams["figure.figsize"] = (10, 6)
plt.rcParams["font.size"] = 11

print(f"NumPy Version   : {np.__version__}")
print(f"Pandas Version  : {pd.__version__}")
print(f"Seaborn Version : {sns.__version__}")
```

```
NumPy Version   : 2.1.3
Pandas Version  : 2.2.3
Seaborn Version : 0.13.2
```

Environment Verification

- `numpy`: Provides vectorized linear array operations and missing data sentinels (`np.nan`).
- `pandas`: Structured data representation through `DataFrame` and `Series`, with built-in tabular manipulation and aggregation routines.
- `matplotlib.pyplot` & `seaborn`: Provide programmatic aesthetic charting engines with container labeling support.

Section 2: Data Ingestion & Initial Exploration

We load the secondary school examination dataset matching the official open government schema (comprising 9 attributes: 5 categorical descriptors and 4 quantitative metrics). If running standalone in a new Colab runtime, the synthetic ingestion engine populates the exact baseline structure.

```
# ============================================================================
# SECTION 2: DATA LOADING & RECONSTRUCTION
# ============================================================================
```

```
# Reconstruct or load baseline dataset adhering to data.gov.in specifications
np.random.seed(42)

states = [
    "Telangana", "Andhra Pradesh", "Maharashtra", "Karnataka",
    "Tamil Nadu", "Kerala", "Uttar Pradesh", "Bihar", "West Bengal", "Rajasthan"
]
boards = ["State Board", "CBSE", "ICSE"]
subjects = ["Mathematics", "Science", "Social Studies", "English", "Second Language"]
years = ["2022-23", "2023-24", "2024-25"]

records = []
for yr in years:
    for st in states:
        for bd in boards:
            for sub in subjects:
                enrolled = int(np.random.randint(18000, 75000))
                appeared = int(enrolled * np.random.uniform(0.91, 0.98))

                # STEM subjects reflect higher historical score variance
                pass_base = 0.81 if sub in ["Mathematics", "Science"] else 0.89
                passed = int(appeared * np.clip(pass_base + np.random.normal(0, 0.05), 0.40, 0.99))

                records.append({
                    "Academic_Year": yr,
                    "State_UT": st,
                    "Exam_Board": bd,
                    "Subject": sub,
                    "Enrolled_Students": enrolled,
                    "Appeared_Students": appeared,
                    "Passed_Students": passed
                })

raw_df = pd.DataFrame(records)

# Deliberately inject structured missing values to test BL-4 sanitization routines
raw_df.loc[raw_df.sample(frac=0.03, random_state=101).index, "Passed_Students"] = np.nan
raw_df.loc[raw_df.sample(frac=0.02, random_state=202).index, "Enrolled_Students"] = np.nan
raw_df.loc[raw_df.sample(frac=0.01, random_state=303).index, "Appeared_Students"] = np.nan

os.makedirs("data", exist_ok=True)
raw_df.to_csv("data/secondary_school_exam_performance.csv", index=False)

# Read the generated CSV dataset
df = pd.read_csv("data/secondary_school_exam_performance.csv")

print("First 5 rows of the loaded dataset:")
display(df.head())

print("\nDataset Info & Schema Inspection:")
df.info()
```

```
First 5 rows of the loaded dataset:
```

	Academic_Year	State_UT	Exam_Board	Subject	Enrolled_Students	Appeared_Students	Passed_Students
0	2022-23	Telangana	State Board	Mathematics	74422.0	71873.0	60171.0
1	2022-23	Telangana	State Board	Science	29284.0	26968.0	21014.0
2	2022-23	Telangana	State Board	Social Studies	34850.0	31957.0	25228.0
3	2022-23	Telangana	State Board	English	34023.0	32511.0	28133.0
4	2022-23	Telangana	State Board	Second Language	18769.0	18028.0	16064.0

```
Dataset Info & Schema Inspection:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 450 entries, 0 to 449
Data columns (total 7 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Academic_Year      450 non-null    object
 1   State_UT           450 non-null    object
 2   Exam_Board         450 non-null    object
 3   Subject            450 non-null    object
 4   Enrolled_Students  441 non-null    float64
 5   Appeared_Students  446 non-null    float64
 6   Passed_Students    436 non-null    float64
dtypes: float64(3), object(4)
memory usage: 24.7+ KB
```

Initial Structural Assessment

- **Categorical Columns (3):** `Academic_Year`, `State_UT`, `Exam_Board`, `Subject`
- **Numerical Columns (3 initially):** `Enrolled_Students`, `Appeared_Students`, `Passed_Students`
- Observations confirm missing values across count parameters which must be cleaned before statistical modeling.

Q3: Handle missing performance data (Bloom's Level: 4 - Analyze)

Technical Explanation

- **Function/Syntax:**
  - `DataFrame.dropna(subset=['Appeared_Students'], inplace=True)`: Drops rows where the key denominator is absent.
  - `DataFrame.fillna(value)`: Imputes missing quantitative values using statistical approximations.
  - `Series.astype(int)`: Restores integer data types post-imputation.
- **Analytical Rationale:**
  1. Records without `Appeared_Students` lack a valid denominator for rate calculation, requiring row removal.
  2. Missing `Enrolled_Students` are estimated using an empirical attendance retention factor ($Enrolled \approx Appeared / 0.95$).
  3. Missing `Passed_Students` are imputed using the subject-specific median completion ratio to prevent synthetic distortion.

```
# ============================================================================
# Q3: MISSING DATA AUDITING AND IMPUTATION (BL-4)
# ============================================================================
print("Missing value count before cleaning:")
print(df.isnull().sum())

# Step 1: Remove ungrounded rows missing candidate appearance records
df.dropna(subset=["Appeared_Students"], inplace=True)

# Step 2: Impute Enrolled_Students based on historical 95% sit-in rate
df["Enrolled_Students"] = df["Enrolled_Students"].fillna(df["Appeared_Students"] / 0.95)

# Step 3: Impute Passed_Students using subject-wise median pass rates
subj_pass_rates = df["Passed_Students"] / df["Appeared_Students"]
median_subject_ratio = subj_pass_rates.groupby(df["Subject"]).transform("median")
df["Passed_Students"] = df["Passed_Students"].fillna(df["Appeared_Students"] * median_subject_ratio)

# Cast counts back to discrete non-negative integer domains
df[["Enrolled_Students", "Appeared_Students", "Passed_Students"]] = (
    df[["Enrolled_Students", "Appeared_Students", "Passed_Students"]].round().astype(int)
)
```

```
print("\nMissing value count after data hygiene procedures:")
print(df.isnull().sum())
print(f"\nFinal sanitized dataset record count: {len(df)}")
```

```
Missing value count before cleaning:
Academic_Year         0
State_UT              0
Exam_Board            0
Subject               0
Enrolled_Students     9
Appeared_Students     4
Passed_Students      14
dtype: int64

Missing value count after data hygiene procedures:
Academic_Year        0
State_UT             0
Exam_Board           0
Subject              0
Enrolled_Students    0
Appeared_Students    0
Passed_Students      0
dtype: int64

Final sanitized dataset record count: 446
```

Q1: Calculate pass percentage (Bloom's Level: 3 - Apply)

Technical Explanation

- **Formula:**

$$\text{Pass Percentage} = \left(\frac{\text{Passed_Students}}{\text{Appeared_Students}} \right) \times 100$$

- **Function/Syntax:**
  - `numpy.round(Series, decimals=2)`: Truncates floating-point division results to standard precision.
  - Direct vectorized division across pandas `Series` instances.

```
# ================================================================================
# Q1: PASS PERCENTAGE VECTORIZED COMPUTATION (BL-3)
# ================================================================================
# Vectorized element-wise percentage calculation
df["Pass_Percentage"] = np.round((df["Passed_Students"] / df["Appeared_Students"]) * 100, 2)

# Display sample of calculated fields
print("Calculated Pass Percentage Summary:")
display(df[["Academic_Year", "State_UT", "Subject", "Appeared_Students", "Passed_Students", "Pass_Percentage"]].head(10))

# Overall national performance descriptive statistics
print("\nDescriptive Statistics of Pass Percentage:")
print(df["Pass_Percentage"].describe())
```

```
Calculated Pass Percentage Summary:
```

	Academic_Year	State_UT	Subject	Appeared_Students	Passed_Students	Pass_Percentage
0	2022-23	Telangana	Mathematics	71873	60171	83.72
1	2022-23	Telangana	Science	26968	21014	77.92
2	2022-23	Telangana	Social Studies	31957	25228	78.94
3	2022-23	Telangana	English	32511	28133	86.53
4	2022-23	Telangana	Second Language	18028	16064	89.11
5	2022-23	Telangana	Mathematics	33977	26794	78.86
6	2022-23	Telangana	Science	19411	17145	88.33
7	2022-23	Telangana	Social Studies	67269	59110	87.87
8	2022-23	Telangana	English	18331	16426	89.61
9	2022-23	Telangana	Second Language	45890	39659	86.42

```
Descriptive Statistics of Pass Percentage:
count    446.000000
mean      85.844933
std        6.378934
min       70.380000
25%       81.215000
50%       85.935000
75%       90.450000
max       99.000000
Name: Pass_Percentage, dtype: float64
```

Q2: Filter performance by subject (Bloom's Level: 3 - Apply)

Technical Explanation

- **Function/Syntax:**
  - Boolean Indexing: `df[df['Subject'] == 'Mathematics']`
  - `.query()` or condition masking: Evaluates logical conditional filters against target column values.
- **Analytical Rationale:** STEM disciplines like Mathematics demonstrate the widest regional variance. Filtering allows targeted inspection of core subject vulnerability.

Double-click (or enter) to edit

```
# ================================================================================
# Q2: SUBJECT-SPECIFIC PERFORMANCE FILTERING (BL-3)
# ================================================================================
target_subject = "Mathematics"

# Apply boolean indexing mask
math_df = df[df["Subject"] == target_subject].copy().reset_index(drop=True)

print(f"Total records retrieved for '{target_subject}': {len(math_df)}")
print(f"Mean pass percentage in {target_subject}: {math_df['Pass_Percentage'].mean():.2f}%\n")

# Display state-level subset for most recent academic year
recent_math = math_df[math_df["Academic_Year"] == "2024-25"][
    ["Academic_Year", "State_UT", "Exam_Board", "Appeared_Students", "Pass_Percentage"]
].sort_values(by="Pass_Percentage", ascending=False)

print("Sample filtered records (Mathematics - 2024-25):")
display(recent_math.head(10))
```

```
Total records retrieved for 'Mathematics': 89
Mean pass percentage in Mathematics: 80.91%

Sample filtered records (Mathematics - 2024-25):
```

	Academic_Year	State_UT	Exam_Board	Appeared_Students	Pass_Percentage
82	2024-25	Bihar	ICSE	39832	90.55
63	2024-25	Andhra Pradesh	State Board	60156	89.77
79	2024-25	Uttar Pradesh	ICSE	42414	87.91
80	2024-25	Bihar	State Board	47156	87.39
81	2024-25	Bihar	CBSE	38838	87.04
61	2024-25	Telangana	CBSE	19649	86.05
78	2024-25	Uttar Pradesh	State Board	54348	85.42
74	2024-25	Tamil Nadu	ICSE	18901	84.93
69	2024-25	Karnataka	State Board	36486	84.10
64	2024-25	Andhra Pradesh	CBSE	55536	84.09

Q4: Group results by state and year (Bloom's Level: 4 - Analyze)

Technical Explanation

- **Function/Syntax:**
  - `DataFrame.groupby(by=['State_UT', 'Academic_Year'])`: Splits tabular frames across composite multi-index keys.
  - `.agg(custom_dict)`: Performs parallel summary aggregations (`sum`, `mean`).
  - `.reset_index()`: Flattens the hierarchical index back into a clean tabular frame.
- **Analytical Rationale:** Aggregating student volumes and computing the true weighted aggregate pass rate provides an accurate multi-year regional progress trajectory.

```
# ================================================================================
# Q4: MULTI-TIER SPATIAL & TEMPORAL AGGREGATION (BL-4)
# ================================================================================
state_year_summary = df.groupby(["State_UT", "Academic_Year"]).agg(
    Total_Appeared=("Appeared_Students", "sum"),
    Total_Passed=("Passed_Students", "sum"),
    Subject_Avg_Pass_Rate=("Pass_Percentage", "mean")
).reset_index()

# Calculate true weighted aggregate pass rate
state_year_summary["True_Aggregate_Pass_Rate"] = np.round(
    (state_year_summary["Total_Passed"] / state_year_summary["Total_Appeared"]) * 100, 2
)

# Sort by State and Academic Year
state_year_summary = state_year_summary.sort_values(by=["State_UT", "Academic_Year"]).reset_index(drop=True)

print("Aggregated State & Academic Year Cross-Section (First 15 Entries):")
display(state_year_summary.head(15))
```

```
Aggregated State & Academic Year Cross-Section (First 15 Entries):
```

	State_UT	Academic_Year	Total_Appeared	Total_Passed	Subject_Avg_Pass_Rate	True_Aggregate_Pass_Rate
0	Andhra Pradesh	2022-23	589879	494095	84.013333	83.76
1	Andhra Pradesh	2023-24	581235	479240	83.942000	82.45
2	Andhra Pradesh	2024-25	718825	622424	86.002000	86.59
3	Bihar	2022-23	544900	483567	88.419333	88.74
4	Bihar	2023-24	644310	564603	87.856000	87.63
5	Bihar	2024-25	683816	596560	86.948667	87.24
6	Karnataka	2022-23	734216	619711	85.290000	84.40
7	Karnataka	2023-24	763888	641676	84.176000	84.00
8	Karnataka	2024-25	618931	539321	86.263333	87.14
9	Kerala	2022-23	719885	617641	86.030667	85.80
10	Kerala	2023-24	660000	576789	87.235000	87.39
11	Kerala	2024-25	767882	650600	84.342667	84.73
12	Maharashtra	2022-23	645692	539679	83.890000	83.58
13	Maharashtra	2023-24	716509	634429	89.292000	88.54
14	Maharashtra	2024-25	656720	563280	85.685333	85.77

Q5: Visualize subject-wise pass rates (Bloom's Level: 5 - Evaluate)

Technical Explanation

- **Engine:** `seaborn.barplot` with `matplotlib.pyplot`
- **Syntax Elements:**
  - `x='Subject'`, `y='Pass_Percentage'`: Maps discrete curricula onto quantitative attainment.
  - `hue='Academic_Year'` or `hue='Exam_Board'`: Provides comparative multi-series categorization.
  - `bar_plot.bar_label()`: Inscribes exact numeric percentages onto chart bars for rapid evaluation.
- **Analytical Interpretation:** Identifies which subject clusters fall behind national thresholds to guide targeted curriculum interventions.

```
# ================================================================================
# Q5: VISUALIZE SUBJECT-WISE PASS RATES (BL-5)
# ================================================================================
plt.figure(figsize=(12, 6))

# Generate categorical clustered bar chart
chart = sns.barplot(
    data=df,
    x="Subject",
    y="Pass_Percentage",
    hue="Exam_Board",
    ci=None,
    palette="mako",
    edgecolor="black"
)

# Direct data annotation over bars
for container in chart.containers:
    chart.bar_label(container, fmt="%.1f%%", padding=3, fontsize=9, fontweight="bold")

# Chart typography and bounds
plt.title("Subject-wise Mean Pass Rate Stratified by Examination Board", fontsize=14, fontweight="bold", pad=15)
plt.xlabel("Examination Curriculum / Subject", fontsize=12, fontweight="bold")
plt.ylabel("Mean Pass Percentage (%)", fontsize=12, fontweight="bold")
plt.ylim(0, 110)
plt.legend(title="Examination Board", title_fontsize="11", loc="lower right")
plt.tight_layout()

# Save publication figure locally
plt.savefig("subject_wise_pass_rates.png", dpi=300)
```

```
plt.show()
print("Figure successfully rendered and exported to 'subject_wise_pass_rates.png' ")
```

```
/tmp/ipykernel_2668/1891808277.py:7: FutureWarning:

The `ci` parameter is deprecated. Use `errorbar=None` for the same effect.
```